ENGINE CONTROL SYSTEM [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0140q1139900

Outline

- L-jetronic*1 and D-jetronic*2 type detectors have been combined for intake air amount detection, improving the accuracy of the intake air amount measurement.
  - — MAF sensor adopted
  - — MAP sensor adopted
  - — IAT sensor No.1 and No.2 adopted

SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)

- Hydraulic variable valve timing control has been adopted on intake side, improving fuel economy and emission performance.
  - — Intake CMP sensor adopted

SKYACTIV-G 1.5 (with 4-2-1 exhaust system)

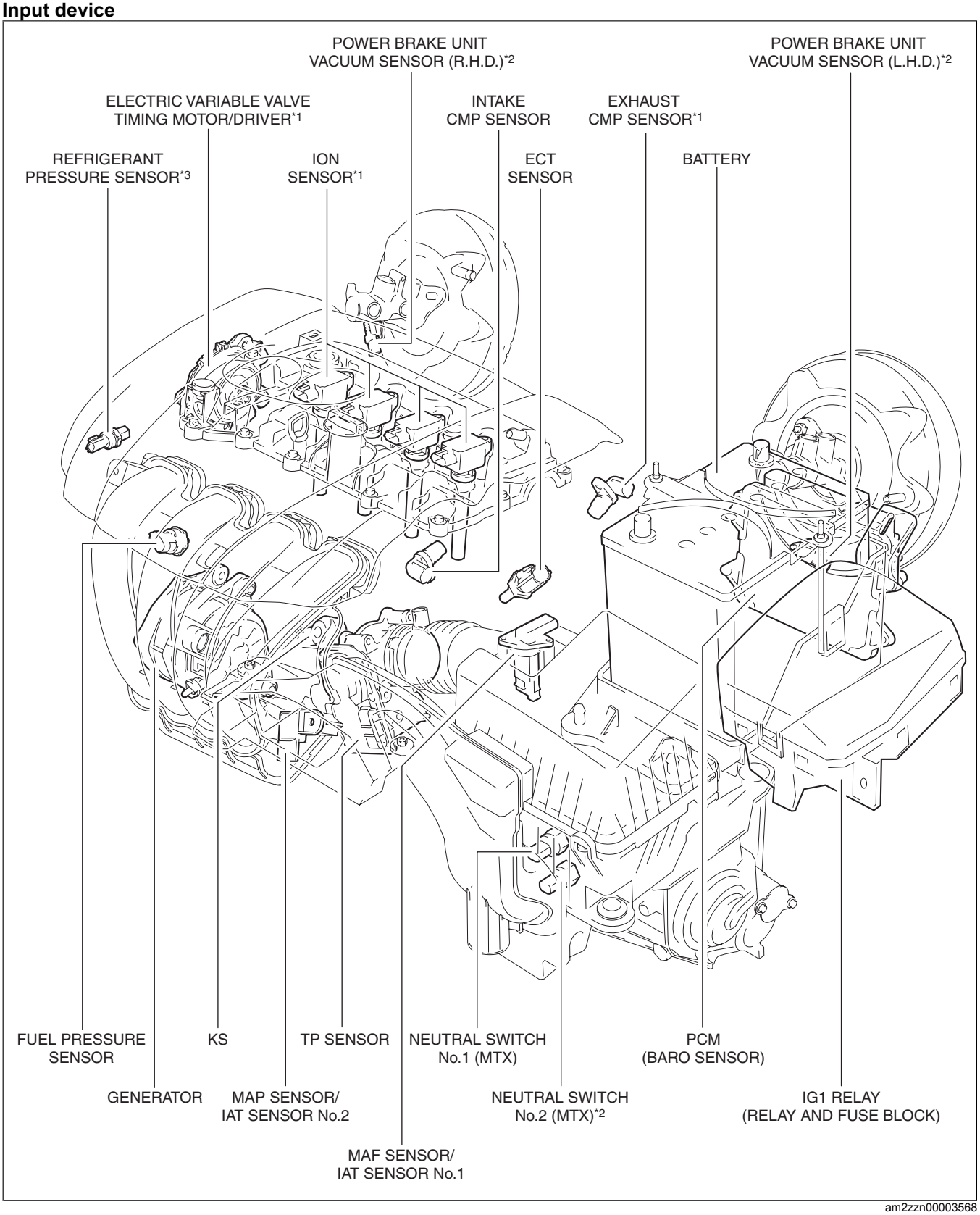
- Valve timing control has been adopted on both sides of the intake and exhaust, improving fuel economy and emission performance.
  Intake side: Electric variable valve timing control
  - — Intake CMP sensor adopted
  - — Electric variable valve timing motor/driver adopted
  - — Electric variable valve timing relay adopted

  Exhaust side: Hydraulic variable valve timing control
  - — Exhaust CMP sensor adopted
- With the adoption of the ion sensor, which detects pre-ignition, engine reliability has been improved.
- Engine oil control has been adopted reducing engine load.
  - — Engine oil solenoid valve adopted
- With the adoption of fuel pump control, fuel pump power consumption has been reduced, improving fuel economy. (with fuel pump control module)
  - — Fuel pump control module adopted
- Generator output control adopted, fuel economy/idling stability improved. (with i-stop system)
  - — Current sensor adopted
- DC-DC converter control has been adopted for improved power supply stability. (without i-ELOOP)
  - — DC-DC converter adopted
- With the adoption of the i-ELOOP, charging efficiency during deceleration is improved. Because loss of engine force does not occur when the battery is recharged during deceleration, fuel economy is improved. (with i-ELOOP)
- LIN communication has been adopted to the current sensor (with i-stop system) and DC-DC converter (without i-ELOOP) for simplified wiring harnesses.

*1 : Measures the intake air amount directly using the MAF sensor.

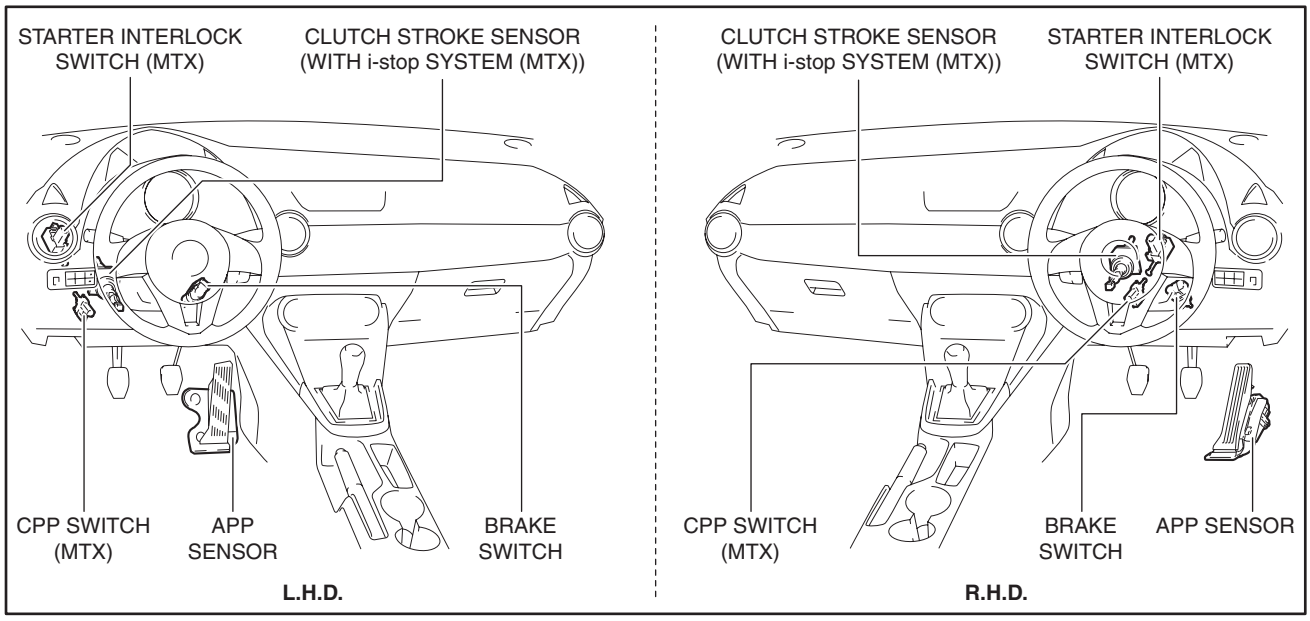
*2 : Measures the intake air pressure introduced into the cylinder using the MAP sensor and calculates the intake air amount indirectly.

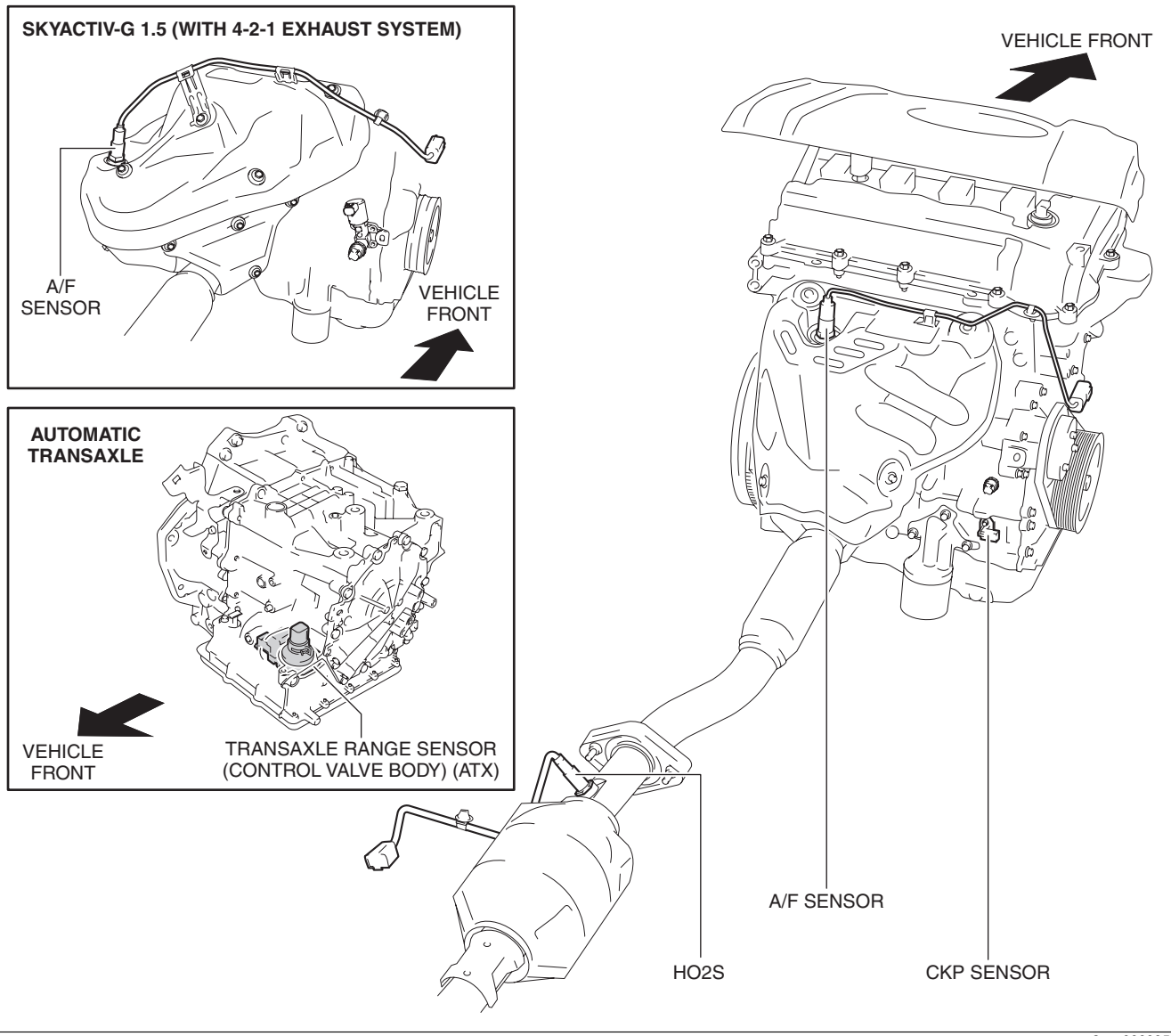
Structural View
Input device



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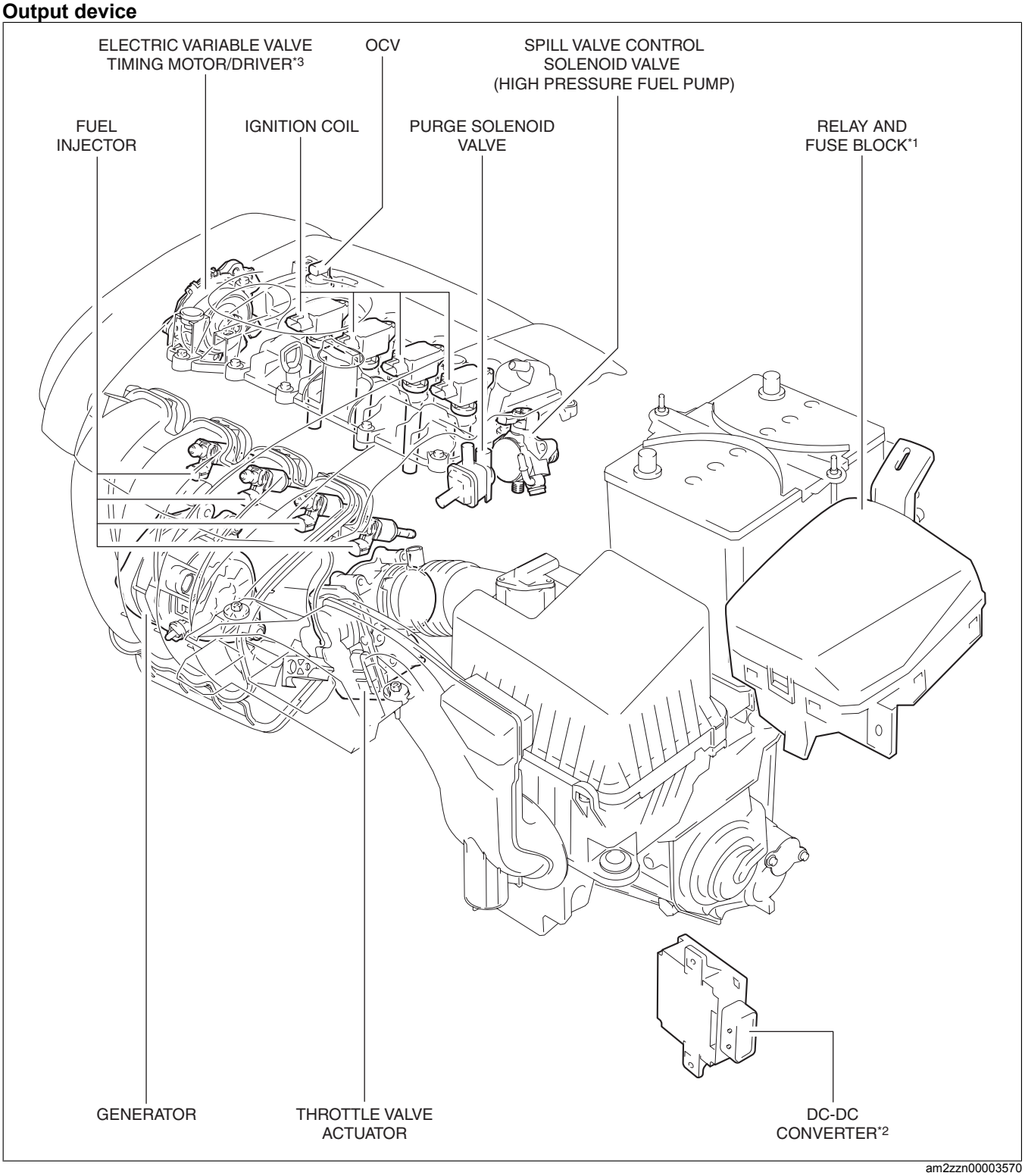
*1 : SKYACTIV-G 1.5 (with 4-2-1 exhaust system)
*2 : With i-stop system
*3 : With air conditioner





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Output device

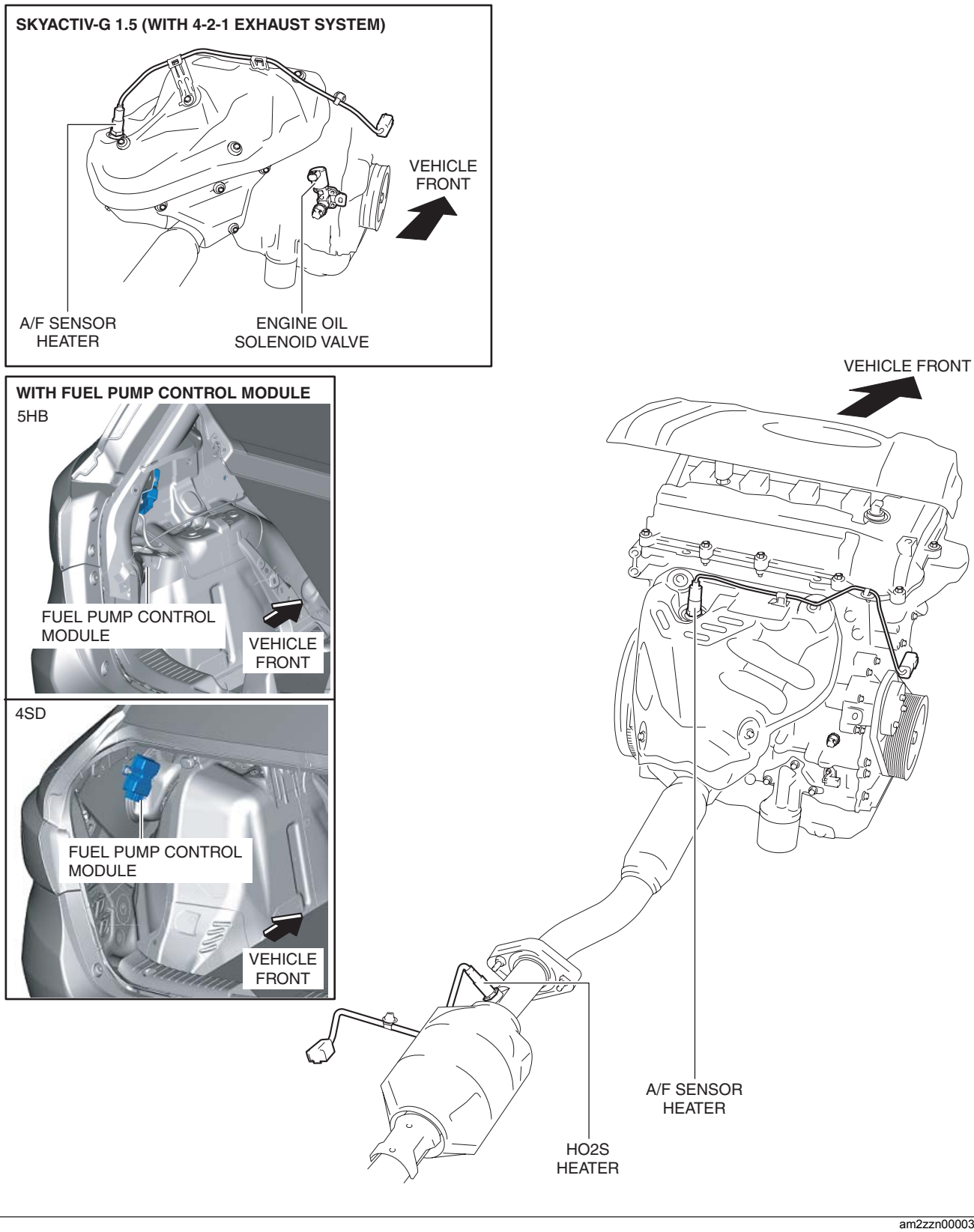


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*1 : Main relay, fuel injector relay, fuel pump relay, A/C relay, cooling fan relay No.1, No.2, starter relay, electric variable valve timing relay

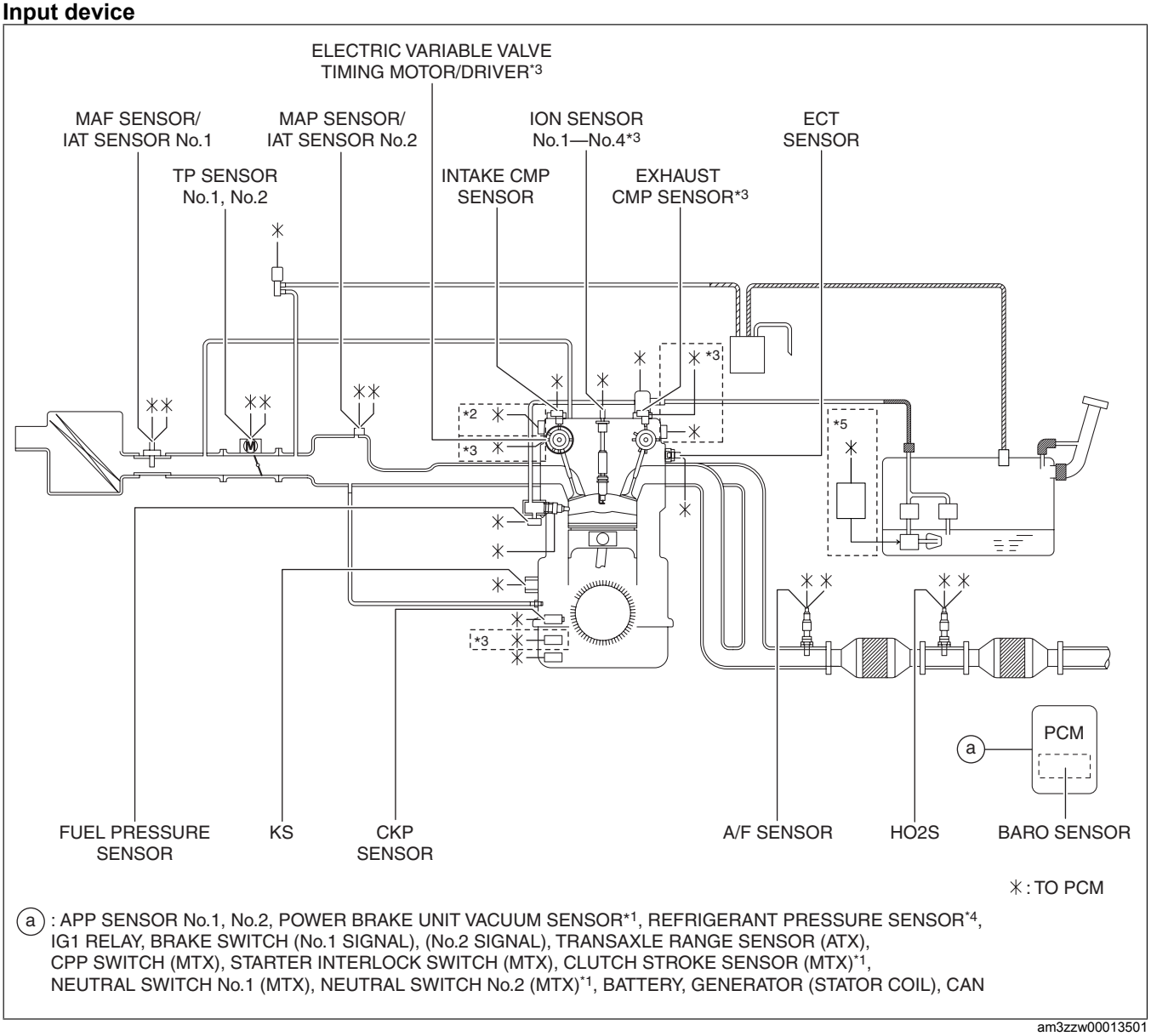
*2 : With i-stop system, without i-ELOOP

*3 : SKYACTIV-G 1.5 (with 4-2-1 exhaust system)



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System Diagram
Input device

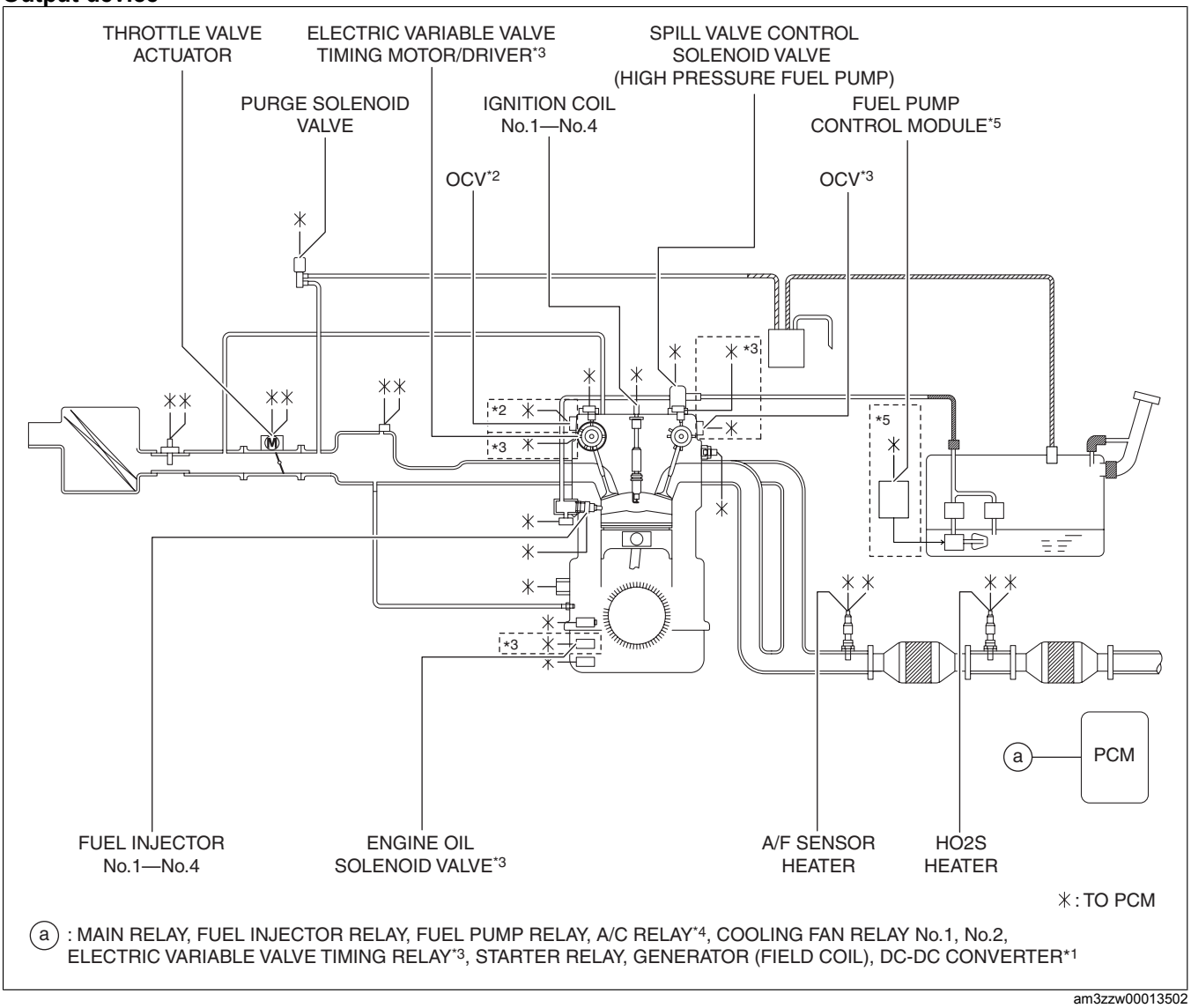


(a) : APP SENSOR No.1, No.2, POWER BRAKE UNIT VACUUM SENSOR*1, REFRIGERANT PRESSURE SENSOR*4, IG1 RELAY, BRAKE SWITCH (No.1 SIGNAL), (No.2 SIGNAL), TRANSAXLE RANGE SENSOR (ATX), CPP SWITCH (MTX), STARTER INTERLOCK SWITCH (MTX), CLUTCH STROKE SENSOR (MTX)*1, NEUTRAL SWITCH No.1 (MTX), NEUTRAL SWITCH No.2 (MTX)*1, BATTERY, GENERATOR (STATOR COIL), CAN

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*1 : With i-stop system
*2 : SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)
*3 : SKYACTIV-G 1.5 (with 4-2-1 exhaust system)
*4 : With air conditioner
*5 : With fuel pump control module

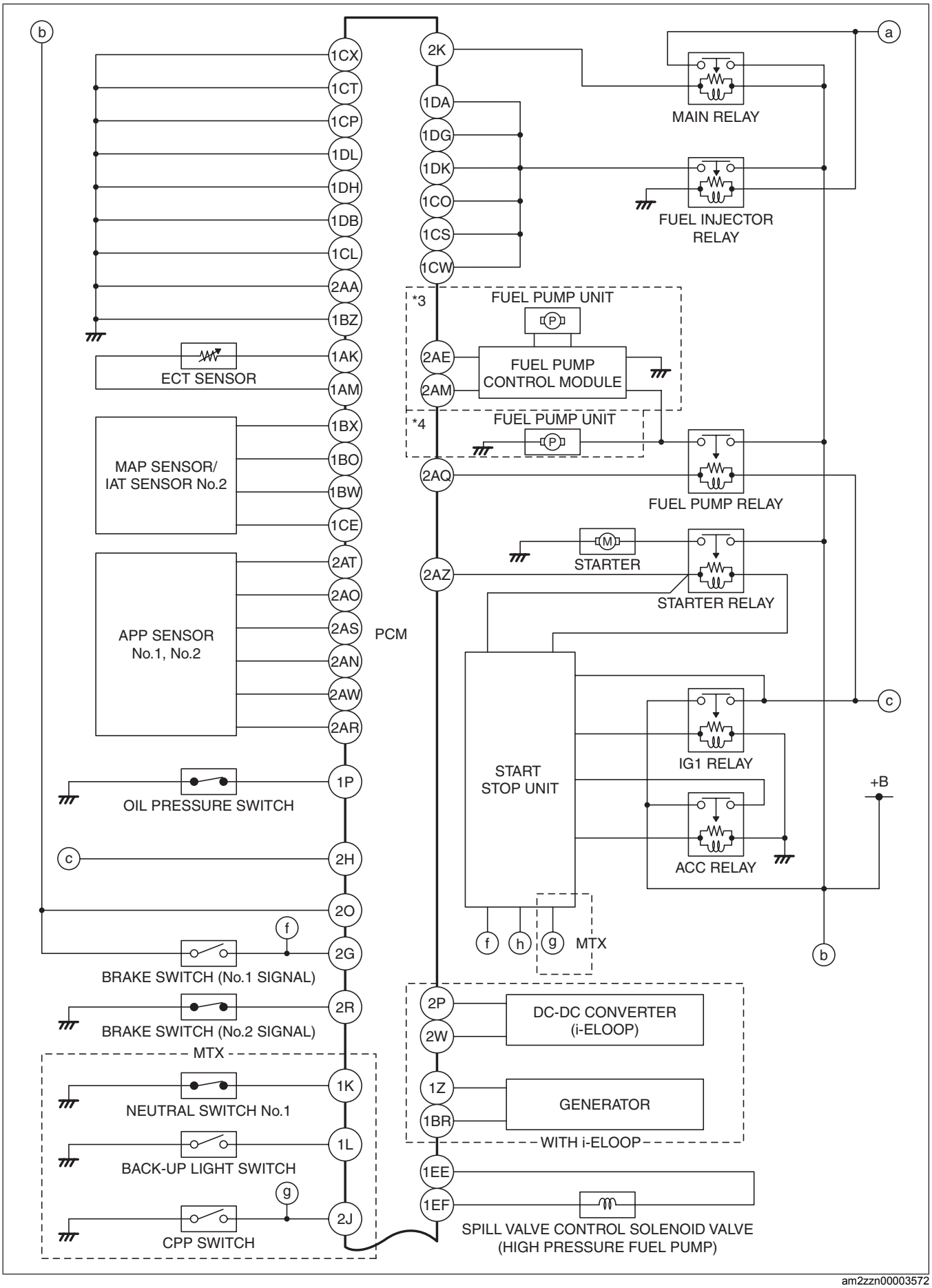
Output device



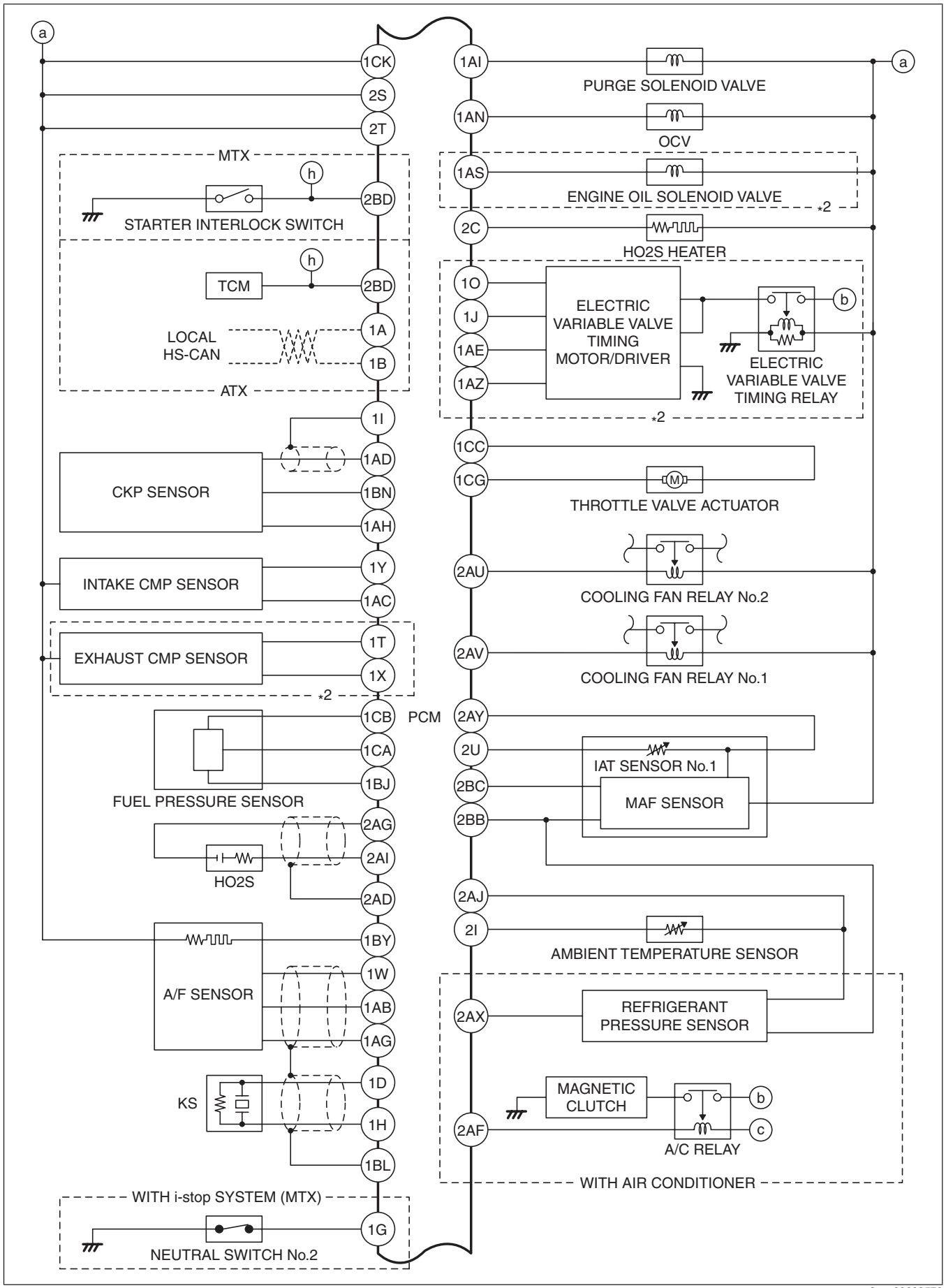
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*1 : With i-stop system, without i-ELOOP
*2 : SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)
*3 : SKYACTIV-G 1.5 (with 4-2-1 exhaust system)
*4 : With air conditioner
*5 : With fuel pump control module

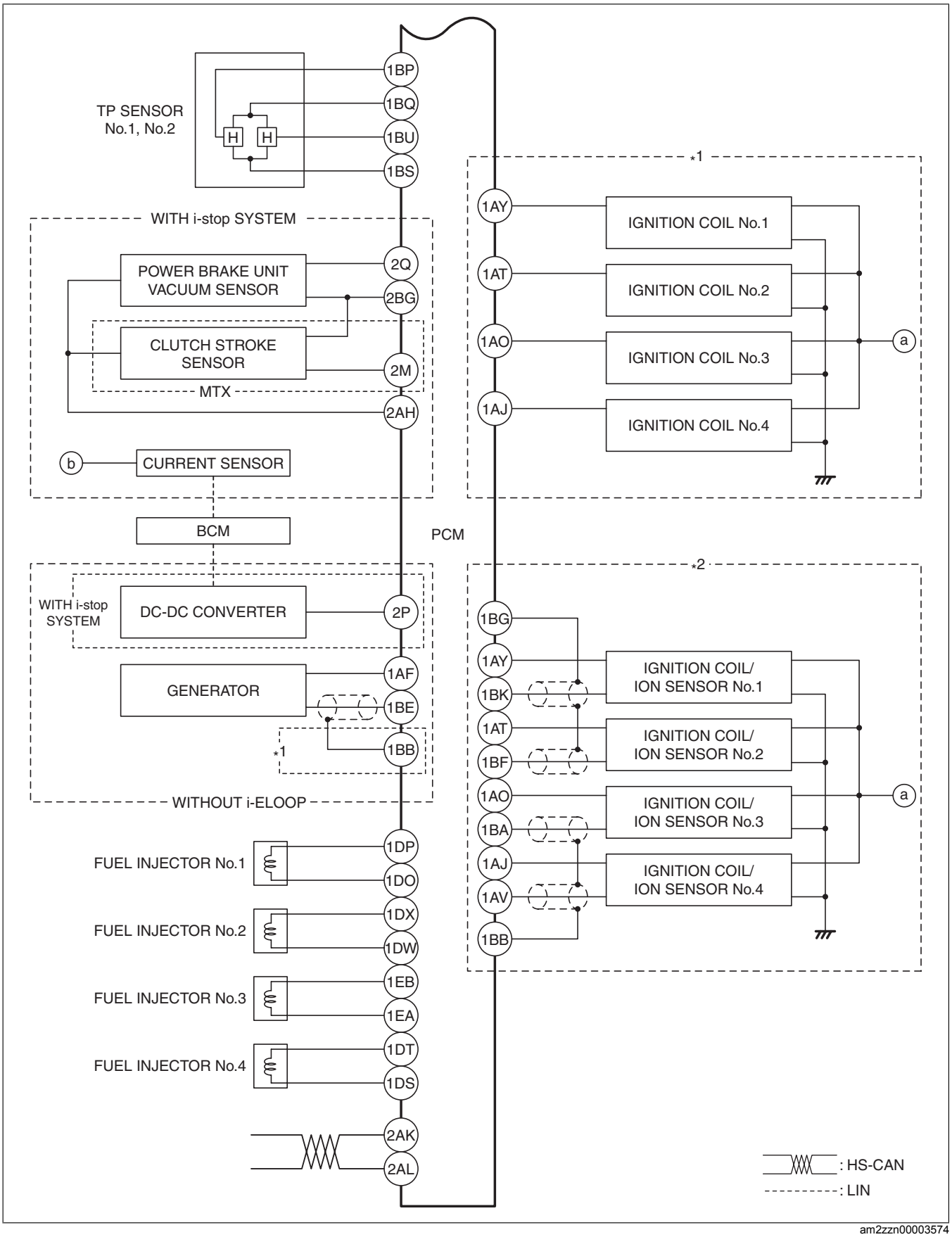
System Wiring Diagram



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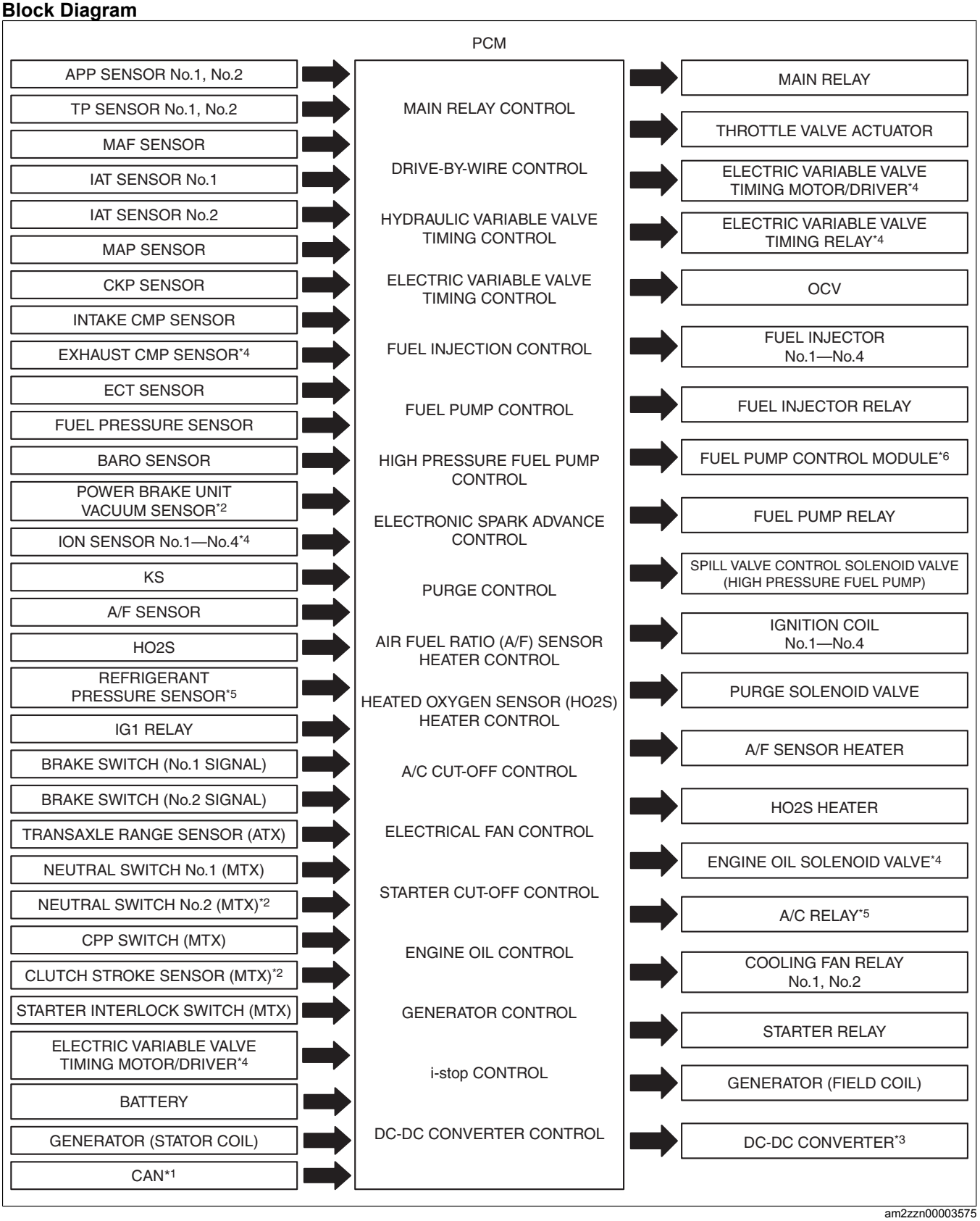
*1 : SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1exhaust system)

*2 : SKYACTIV-G 1.5 (with 4-2-1exhaust system)

*3 : With fuel pump control module

*4 : Without fuel pump control module

Block Diagram



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*1 : DSC HU/CM, TCM (ATX), EPS control module, instrument cluster, BCM, start stop unit

*2 : With i-stop system

*3 : With i-stop system, without i-ELOOP

*4 : SKYACTIV-G 1.5 (with 4-2-1exhaust system)

*5 : With air conditioner

*6 : With fuel pump control module

Relation Chart

- Each control system and their related input and output parts are as follows.

×: Applicable

Item	MAIN RELAY CONTROL	DRIVE-BY-WIRE CONTROL	HYDRAULIC VARIABLE VALVE TIMING CONTROL	ELECTRIC VARIABLE VALVE TIMING CONTROL	FUEL INJECTION CONTROL	FUEL PUMP CONTROL	HIGH PRESSURE FUEL PUMP CONTROL	ELECTRONIC SPARK ADVANCE CONTROL	PURGE CONTROL	A/F SENSOR HEATER CONTROL HO2S HEATER CONTROL	A/C CUT-OFF CONTROL	ELECTRICAL FAN CONTROL	STARTER CUT-OFF CONTROL	GENERATOR CONTROL	i-stop CONTROL	DC-DC CONVERTER CONTROL	ENGINE OIL CONTROL
Input device																	
APP sensor No.1, No.2		×	×		×			×	×		×	×			×		
TP sensor No.1, No.2		×			×						×				×		
MAF sensor		×	×	×	×		×	×	×	×		×			×		×
IAT sensor No.1		×		×	×			×	×			×		×*3	×		
IAT sensor No.2		×		×	×			×							×		
MAP sensor		×		×	×			×	×						×		×
CKP sensor		×	×	×	×	×	×	×	×	×	×	×	×	×	×		×
Intake CMP sensor		×	×*5	×	×			×							×		
Exhaust CMP sensor*6		×	×		×		×	×							×		
ECT sensor		×	×	×	×		×	×	×	×	×	×	×	×*3	×		×
Fuel pressure sensor		×			×	×	×								×		
BARO sensor		×		×	×				×			×			×		
Power brake unit vacuum sensor*2															×		
Ion sensor No.1—No.4*6				×	×												
KS								×									
A/F sensor					×				×								
HO2S					×												
Refrigerant pressure sensor*7											×	×					
IG1 relay	×		×		×	×		×		×		×	×		×	×	
Brake switch (No.1 signal)		×													×		
Brake switch (No.2 signal)		×															
Transaxle range sensor (ATX)		×			×								×		×		
CPP switch (MTX)		×			×			×			×				×		
Starter interlock switch (MTX)													×				
Clutch stroke sensor (MTX) *2															×		
Neutral switch No.1 (MTX)		×			×			×			×				×		
Neutral switch No.2 (MTX)*2															×		
Electric variable valve timing motor/driver*6				×											×		
Battery			×			×	×	×	×					×	×		
Generator (Stator coil)		×															
CAN*1		×			×			×			×	×	×	×*2	×	×	
Output device																	
Main relay	×																
Throttle valve actuator		×													×		

Item	MAIN RELAY CONTROL	DRIVE-BY-WIRE CONTROL	HYDRAULIC VARIABLE VALVE TIMING CONTROL	ELECTRIC VARIABLE VALVE TIMING CONTROL	FUEL INJECTION CONTROL	FUEL PUMP CONTROL	HIGH PRESSURE FUEL PUMP CONTROL	ELECTRONIC SPARK ADVANCE CONTROL	PURGE CONTROL	A/F SENSOR HEATER CONTROL HO2S HEATER CONTROL	A/C CUT-OFF CONTROL	ELECTRICAL FAN CONTROL	STARTER CUT-OFF CONTROL	GENERATOR CONTROL	i-stop CONTROL	DC-DC CONVERTER CONTROL	ENGINE OIL CONTROL
Electric variable valve timing motor/driver*6				×	×										×		
Electric variable valve timing relay*6				×											×		
OCV			×														
Fuel injector No.1—No.4					×										×		
Fuel injector relay					×										×		
Fuel pump control module*8						×											
Fuel pump relay						×											
Spill valve control solenoid valve (High pressure fuel pump)							×										
Ignition coil No.1—No.4								×							×		
Purge solenoid valve									×								
A/F sensor heater										×							
HO2S heater										×							
A/C relay*7											×						
Cooling fan relay No.1, No.2												×					
Starter relay													×		×		
Generator (Field coil)														×	×		
DC-DC converter*4															×	×	
Engine oil solenoid valve*6																	×

*1 : TCM (ATX), DSC HU/CM or ABS HU/CM, EPS control module, instrument cluster, BCM, start stop unit
*2 : With i-stop system
*3 : Without i-stop system
*4 : With i-stop system, without i-ELOOP
*5 : SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1exhaust system)
*6 : SKYACTIV-G 1.5 (with 4-2-1exhaust system)
*7 : With air conditioner
*8 : With fuel pump control module